DATABASE TECHNOLOGIES – ASSIGNMENT 10

PL-SQL

1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, Perimeter = 2*(L+W) and Area = L*W.

```
DELIMITER $$


CREATE PROCEDURE sp_areaAndPerimeter(IN L int,IN W int,OUT area INT,OUT peri INT )

BEGIN


SET peri = 2*(L+W);

SET area = L*W;


END;

$$

DELIMITER ;



CALL sp_areaAndPerimeter(6,4,@a,@p);

SELECT @a AS AREA,@p AS PERIMETER

FROM DUAL;
```

```
[d5_96525_Deven@classwork]>DELIMITER $$
[d5_96525_Deven@classwork]>
[d5_96525_Deven@classwork]>CREATE PROCEDURE sp_areaAndPerimeter(IN L int,IN W int,OUT area INT,OUT peri INT )
    -> BEGIN
    ->
    -> SET peri = 2*(L+W);
    -> SET area = L*W;
    ->
    -> END;
    -> $$
Query OK, 0 rows affected (0.172 sec)

[d5_96525_Deven@classwork]>DELIMITER ;
[d5_96525_Deven@classwork]>
[d5_96525_Deven@classwork]>
[d5_96525_Deven@classwork]>CALL sp_areaAndPerimeter(6,4,@a,@p);
Query OK, 0 rows affected (0.004 sec)

[d5_96525_Deven@classwork]>SELECT @a AS AREA,@p AS PERIMETER
    -> FROM DUAL;
+------+-----------+
| AREA | PERIMETER |
+------+-----------+
|   24 |        20 |
+------+-----------+
1 row in set (0.003 sec)

[d5_96525_Deven@classwork]>
```

2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```
DELIMITER $$


CREATE PROCEDURE sp_square_cube(num INT)

BEGIN


DECLARE temp1 INT;

DECLARE temp2 INT;




SET temp1 = num * num;

INSERT INTO results VALUE (temp1,CONCAT('Square of ',num));


SET temp2 = num * num *num;

INSERT INTO results VALUE (temp2,CONCAT('Cube of ',num));


END;

$$

DELIMITER ;
```

TRUNCATE results;

CALL sp_square_cube(4);

SELECT *

FROM results;

```
[d5_96525_Deven@classwork]>DELIMITER $$
[d5_96525_Deven@classwork]>
[d5_96525_Deven@classwork]>CREATE PROCEDURE sp_square_cube(num INT)
    -> BEGIN
    ->
    -> DECLARE temp1 INT;
    -> DECLARE temp2 INT;
    ->
    ->
    -> SET temp1 = num * num;
    -> INSERT INTO results VALUE (temp1,CONCAT('Square of ',num));
    ->
    -> SET temp2 = num * num *num;
    -> INSERT INTO results VALUE (temp2,CONCAT('Cube of ',num));
    ->
    -> END;
    -> $$
Query OK, 0 rows affected (0.155 sec)

[d5_96525_Deven@classwork]>DELIMITER ;
[d5_96525_Deven@classwork]>TRUNCATE results;
Query OK, 0 rows affected (0.420 sec)

[d5_96525_Deven@classwork]>CALL sp_square_cube(4);
Query OK, 1 row affected (0.066 sec)

[d5_96525_Deven@classwork]>SELECT *
    -> FROM results;
+-------+-------------+
| value | description |
+-------+-------------+
|    16 | Square of 4 |
|    64 | Cube of 4   |
+-------+-------------+
2 rows in set (0.005 sec)

[d5_96525_Deven@classwork]>
```

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are

C= (F-32) *5/9

F= 9/5*C + 32

```
[d5_96525_Deven@classwork]>DELIMITER $$
[d5_96525_Deven@classwork]>
[d5_96525_Deven@classwork]>CREATE PROCEDURE sp_convert(IN T FLOAT, IN ForC CHAR(1), OUT temp FLOAT)
    -> BEGIN
    ->     IF ForC = 'F' THEN
    ->         SET temp = (T - 32) * (5/9);
    ->     ELSE
    ->         SET temp = (T * 9/5) + 32;
    ->     END IF;
    -> END;
    -> $$
Query OK, 0 rows affected (0.164 sec)

[d5_96525_Deven@classwork]>
[d5_96525_Deven@classwork]>DELIMITER ;
[d5_96525_Deven@classwork]>CALL sp_convert(80, 'F', @temp);
Query OK, 0 rows affected (0.007 sec)

[d5_96525_Deven@classwork]>SELECT @temp;
+--------------------+
| @temp              |
+--------------------+
| 26.668800354003906 |
+--------------------+
1 row in set (0.002 sec)

[d5_96525_Deven@classwork]>
```

```
DELIMITER $$


CREATE PROCEDURE sp_convert(IN T FLOAT, IN ForC CHAR(1), OUT temp FLOAT)

BEGIN

   IF ForC = 'F' THEN

      SET temp = (T - 32) * (5/9);

   ELSE

      SET temp = (T * 9/5) + 32;

   END IF;

END;

$$


DELIMITER ;




CALL sp_convert(80, 'F', @temp);
```

SELECT @temp;

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```
[d5_96525_Deven@classwork]>DELIMITER $$
[d5_96525_Deven@classwork]>
[d5_96525_Deven@classwork]>CREATE PROCEDURE sp_convertlen(IN inches INT)
    -> BEGIN
    ->
    -> DECLARE yards INT;
    -> DECLARE feet INT;
    -> DECLARE remaining_inches INT;
    ->
    -> SET yards = inches / 36;
    ->
    -> SET feet = (inches % 36) / 12;
    ->
    -> SET remaining_inches = inches % 12;
    ->
    ->
    -> SELECT CONCAT(inches, ' inches equals ', yards, ' yards, ', feet, ' foot, and ', remaining_inches, ' inches') AS Conversion_Result;
    ->
    -> END;
    -> $$
Query OK, 0 rows affected (0.168 sec)

[d5_96525_Deven@classwork]>DELIMITER ;
[d5_96525_Deven@classwork]>CALL sp_convertlen(124);
+-----------------------------------------------+
| Conversion_Result                             |
+-----------------------------------------------+
| 124 inches equals 3 yards, 1 foot, and 4 inches |
+-----------------------------------------------+
1 row in set (0.010 sec)

Query OK, 0 rows affected (0.030 sec)

[d5_96525_Deven@classwork]>
```

DELIMITER $$

CREATE PROCEDURE sp_convertlen(IN inches INT)

BEGIN

DECLARE yards INT;

DECLARE feet INT;

DECLARE remaining_inches INT;

SET yards = inches / 36;

SET feet = (inches % 36) / 12;

SET remaining_inches = inches % 12;

SELECT CONCAT(inches, ' inches equals ', yards, ' yards, ', feet, ' foot, and ', remaining_inches, ' inches') AS Conversion_Result;

```
END;
$$
DELIMITER ;
```

5. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

```
DELIMITER $$

CREATE PROCEDURE sp_above100ornot(IN sp_num1 INT,IN sp_num2 INT)
BEGIN
DECLARE temp INT;
SET temp = sp_num1*sp_num2;

IF temp>100 THEN
        SELECT 'Above 100';
ELSE
        SELECT 'Below 100';
END IF;

END;
$$
DELIMITER ;


CALL sp_above100ornot(20,6);
```

```
[d5_96525_Deven@classwork]>DELIMITER ;
[d5_96525_Deven@classwork]>
[d5_96525_Deven@classwork]>DELIMITER $$
[d5_96525_Deven@classwork]>
[d5_96525_Deven@classwork]>CREATE PROCEDURE sp_above100ornot(IN sp_num1 INT,IN sp_num2 INT)
    -> BEGIN
    -> DECLARE temp INT;
    -> SET temp = sp_num1*sp_num2;
    ->
    -> IF temp>100 THEN
    ->  SELECT 'Above 100';
    -> ELSE
    ->  SELECT 'Below 100';
    -> END IF;
    ->
    -> END;
    -> $$
Query OK, 0 rows affected (0.163 sec)

[d5_96525_Deven@classwork]>DELIMITER ;
[d5_96525_Deven@classwork]>CALL sp_above100ornot(20,6);
+-----------+
| Above 100 |
+-----------+
| Above 100 |
+-----------+
1 row in set (0.124 sec)

Query OK, 0 rows affected (0.142 sec)

[d5_96525_Deven@classwork]>
```